A good real-world example of Azure Resource Manager (ARM) can be seen within Microsoft's own IT organisation. Microsoft's Digital Cloud Engineering Services team had been using JSON-based ARM templates, but these did not always meet the organisation's security requirements. This meant that teams were spending a significant amount of engineering time fixing security issues manually after resources had already been deployed (Microsoft, 2024). This scenario highlights an issue also recognised by the NIST Cybersecurity Framework, where security controls may be in place but are less effective if they are only applied after deployment rather than being built into the process from the start (NIST, no date).

To address this, the team moved from JSON to Bicep and introduced role-based access control for template storage, alongside a CI/CD pipeline using Azure DevOps. This meant that security requirements could be incorporated into the deployment process, rather than relying on engineers to identify and fix problems afterwards (Microsoft, 2024). This approach also supports the shared responsibility principle discussed by Mather, Kumaraswamy and Latif (2009), where both the cloud provider and the organisation using the cloud have responsibilities for maintaining security. Having these controls built into the deployment process makes those responsibilities much easier to manage.

The results were significant. Microsoft reported a 50 per cent reduction in the engineering time required to develop templates, while the time previously spent fixing failed security controls was eliminated. More than 1,100 users also gained easier access to approved templates (Microsoft, 2024). Ni (2024) argues that security improvements are more meaningful when they can be measured rather than simply claimed. In this case, the figures provide clear evidence of how changes to the way infrastructure was designed and deployed improved both security and operational efficiency.

References

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